X-Window & Networking

Goal: Accessing 1of 5 RPi4B from the Over the Internet with graphical displays

Goal: Accessing 1of 5 RPi4B from the Over the Internet with graphical displays

- Examples graphical displays
  - Geany editor
  - Octave matlab like program
  - Gtkwave VCD viewer
  - Mousepad editor
  - GIMP
  - Lazarus IDE (Ulitobo Editor)
    - Bare metal for Pi
      - QEMU, Rpi Zero, RPi3B+, RPi4, and CM4

Internet

RPi3B+ 1Gb
162.197.186.179
pi4-40

pass-thru

ATT Router

RPi4B 4Gb
192.168.1.212
pi4-50

Century Link
192.168.14.
1

RPi4B 4Gb
192.168.1.211
pi4-30

RPi4B 4Gb
192.168.1.245
pi4-60

RPi4B 4Gb
192.168.1.229
pi4-27

RPi5 8Gb
192.168.1.93
pi5-70

RPi3B+ 1Gb
192.168.1.231
pi4-37

Century
Link6
192.168.32.
1

RPi ZeroW
192.168.1.235
wiifiextender

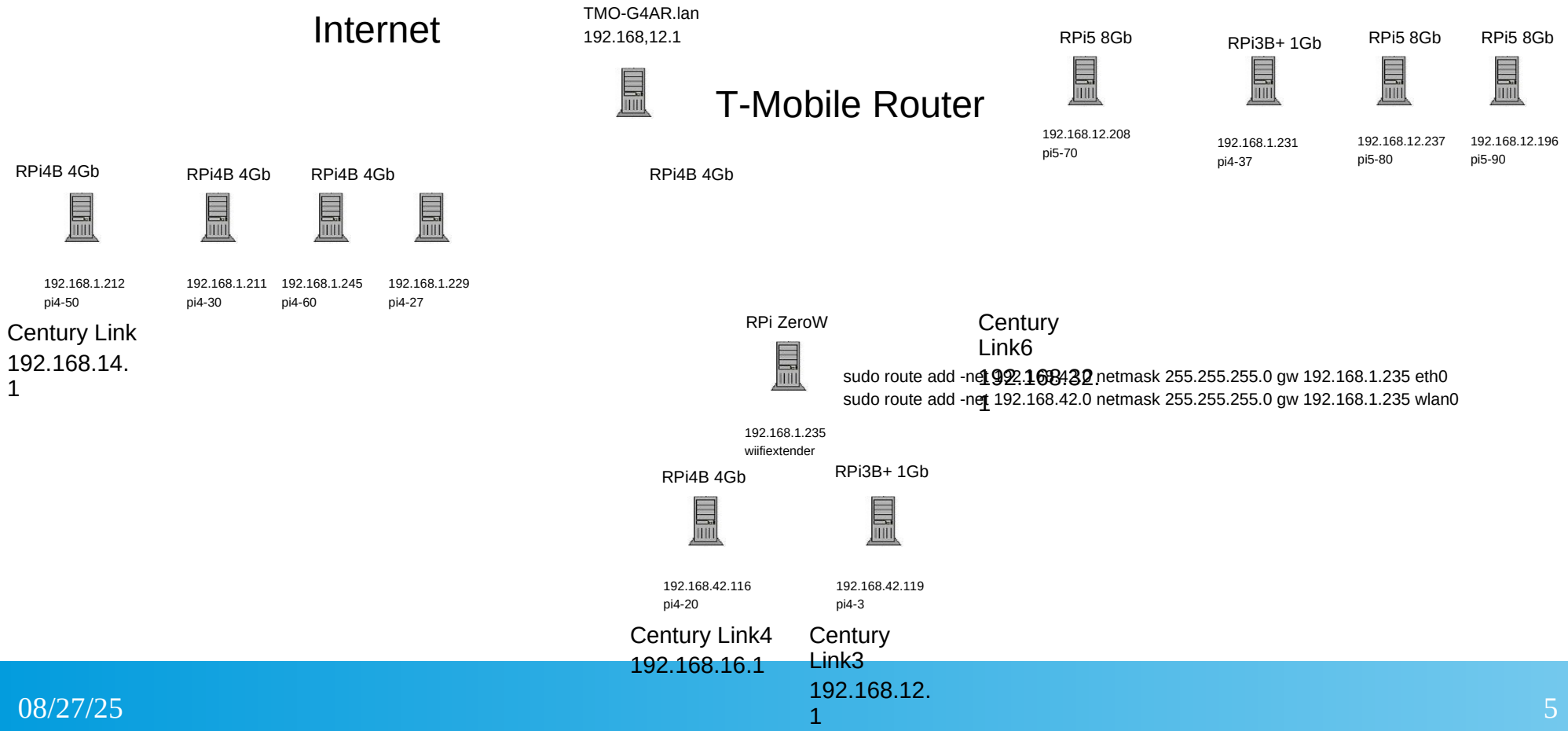
sudo route add -net 192.168.42.0 netmask 255.255.255.0 gw 192.168.1.235 eth0
sudo route add -net 192.168.42.0 netmask 255.255.255.0 gw 192.168.1.235 wlan0

RPi4B 4Gb
192.168.42.116
pi4-20

Century Link4
192.168.16.1

RPi3B+ 1Gb
192.168.42.119
pi4-3

Century
Link3
192.168.12.
1


Internet
TMO-G4AR.lan
192.168,12.1
T-Mobile Router
RPi5 8Gb
192.168.12.208
pi5-70
RPi3B+ 1Gb
192.168.1.231
pi4-37
RPi5 8Gb
192.168.12.237
pi5-80
RPi5 8Gb
192.168.12.196
pi5-90
RPi4B 4Gb
192.168.1.212
pi4-50
RPi4B 4Gb
192.168.1.211
pi4-30
RPi4B 4Gb
192.168.1.245
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pi4-27
Century Link
192.168.14.
1
RPi4B 4Gb
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1
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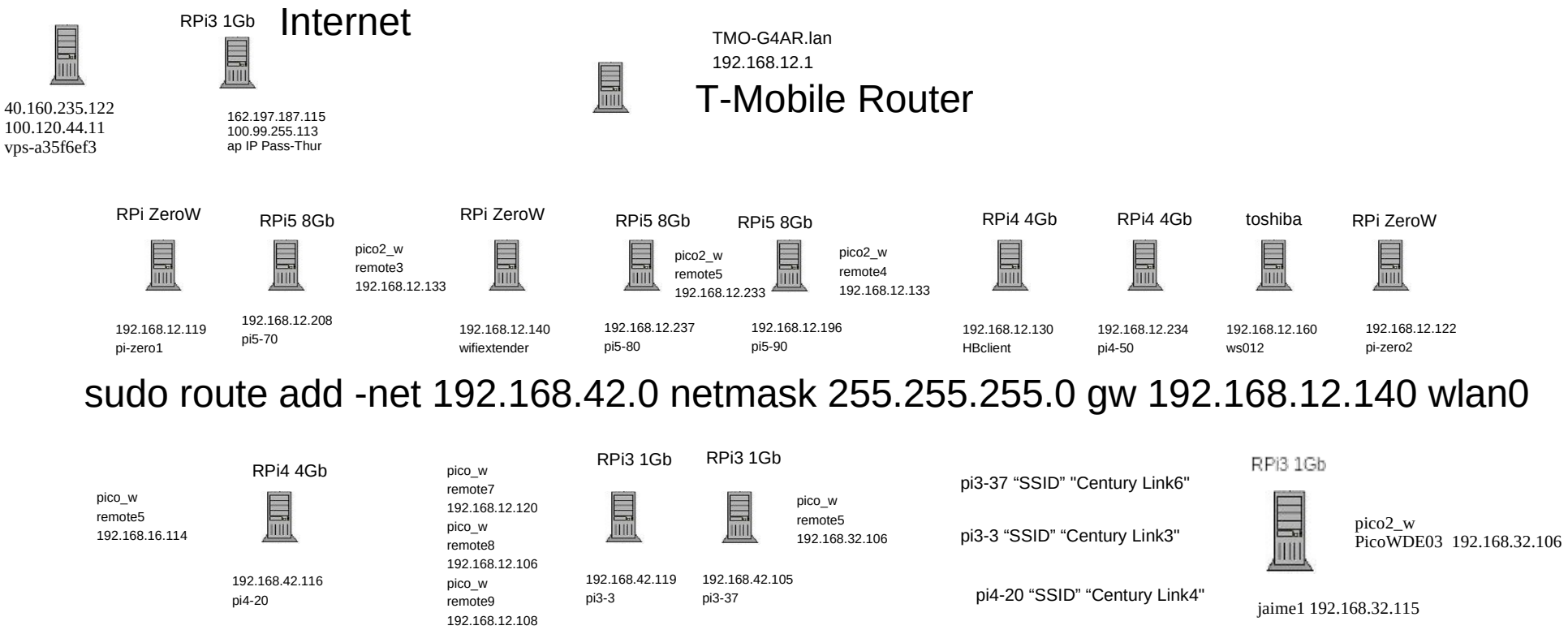
RPi4B 4Gb
192.168.42.116
pi4-20
RPi3B+ 1Gb
192.168.42.119
pi4-3
Century Link4
192.168.16.1
Century
Link3
192.168.12.
1


RPi3 1Gb
Internet
40.160.235.122
100.120.44.11
vps-a35f6ef3
162.197.187.115
100.99.255.113
ap IP Pass-Thur
TMO-G4AR.lan
192.168.12.1
T-Mobile Router
RPi ZeroW
192.168.12.119
pi-zero1
RPi5 8Gb
192.168.12.208
pi5-70
pico2_w
remote3
192.168.12.133
RPi ZeroW
192.168.12.140
wifiextender
RPi5 8Gb
pico2_w
remote5
192.168.12.233
192.168.12.237
pi5-80
RPi5 8Gb
pico2_w
remote4
192.168.12.133
192.168.12.196
pi5-90
RPi4 4Gb
192.168.12.130
HBclient
RPi4 4Gb
192.168.12.234
pi4-50
toshiba
192.168.12.160
ws012
RPi ZeroW
192.168.12.122
pi-zero2
sudo route add -net 192.168.42.0 netmask 255.255.255.0 gw 192.168.12.140 wlan0
pico_w
remote5
192.168.16.114
RPi4 4Gb
192.168.42.116
pi4-20
pico_w
remote7
192.168.12.120
pico_w
remote8
192.168.12.106
pico_w
remote9
192.168.12.108
RPi3 1Gb
192.168.42.119
pi3-3
RPi3 1Gb
pico_w
remote5
192.168.32.106
192.168.42.105
pi3-37
pi3-37 “SSID” "Century Link6"
pi3-3 “SSID” “Century Link3"
pi4-20 “SSID” “Century Link4"
RPi3 1Gb
pico2_w
PicoWDE03 192.168.32.106
jaime1 192.168.32.115

Internet

RPi3B+ 1Gb
162.197.186.179
pi4-40

pass-thru

ATT Router

RPi4B 4Gb
192.168.1.212
pi4-50

RPi4B 4Gb
192.168.1.230
pi4-30

RPi4B 4Gb
192.168.1.218
pi4-2

RPi4B 4Gb
192.168.1.229
pi4-27

RPi5 8Gb
192.168.1.93
pi5-70

RPi5 8Gb
192.168.1.240
pi5-80

RPi5 8Gb
192.168.1.253
pi5-90

Century Link
192.168.14.
1

RPi ZeroW
192.168.1.235
wiifiextender

sudo route add -net 192.168.42.0 netmask 255.255.255.0 gw 192.168.1.235 eth0
sudo route add -net 192.168.42.0 netmask 255.255.255.0 gw 192.168.1.235 wlan0

RPi4B 4Gb
192.168.42.116
pi4-20

RPi3B+ 1Gb
192.168.42.119
pi4-3

RPi3B+ 1Gb
192.168.42.105
pi4-37

Century Link4
192.168.16.1

Century
Link3
192.168.12.
1

Century
Link6
192.168.32.
1

```
devel@pi5-90:~ $ ./scripts-rpi/route/2nd_level-tmo.sh
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         TMO-G4AR.lan    0.0.0.0         UG    600    0        0 wlan0
192.168.12.0    0.0.0.0         255.255.255.0   U     600    0        0 wlan0
192.168.42.0    wifiextender.la 255.255.255.0   UG    0      0        0 wlan0
SIOCADDRT: File exists
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
default         TMO-G4AR.lan    0.0.0.0         UG    600    0        0 wlan0
192.168.12.0    0.0.0.0         255.255.255.0   U     600    0        0 wlan0
192.168.42.0    wifiextender.la 255.255.255.0   UG    0      0        0 wlan0
devel@pi5-90:~ $ traceroute 192.168.42.116
traceroute to 192.168.42.116 (192.168.42.116), 30 hops max, 60 byte
packets
 1  wifiextender.lan (192.168.12.141)  77.323 ms  77.362 ms  77.354 ms
 2  pi4-20 (192.168.42.116)  77.347 ms  77.405 ms  77.399 ms
```

pi4-50
Pico_w Zone1
remote1 192.168.1.160
remote3 192.168.1.178
remote4 192.168.1.177
remote5 192.168.1.168
remote6 192.168.1.175

pi4-37
Pico_w Zone2
SSID Century Link6
remote5 192.168.32.106

pi4-20
Pico_w Zone3
SSID Century Link4
remote5 192.168.16.114

pi4-3
Pico_w Zone4
SSID Century Link3
remote8 192.168.12.120
remote9 192.168.12.108
remote7 192.168.12.106

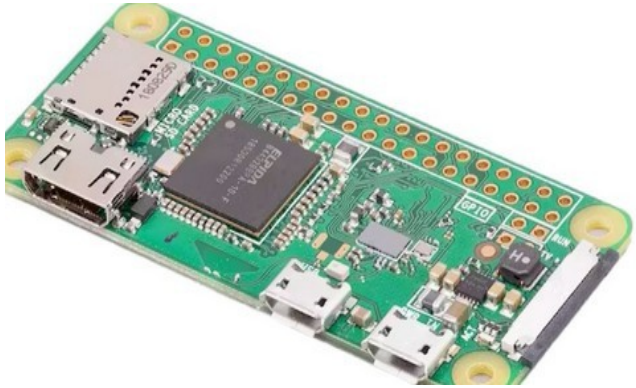
Goal: Accessing 1of 5 RPi4B from the Over the Internet with graphical displays

- Run Graphical Application

- Steps Required
- Need to ssh to
- LAN Home RPi3B+ on 162.197.186.xx

- Need to ssh to 1 of 5 RPi4B

RPi Models

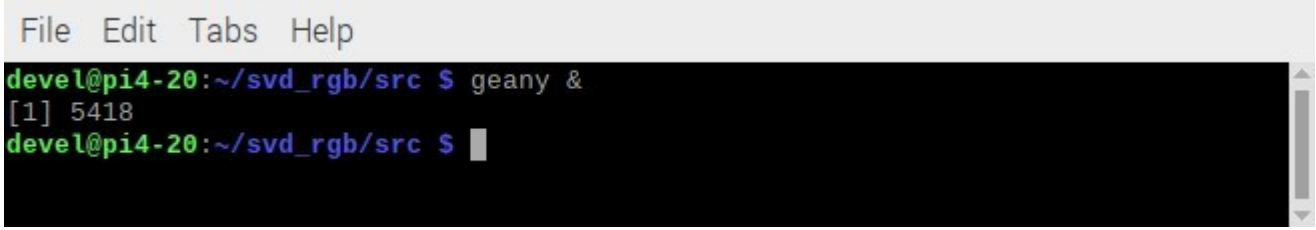


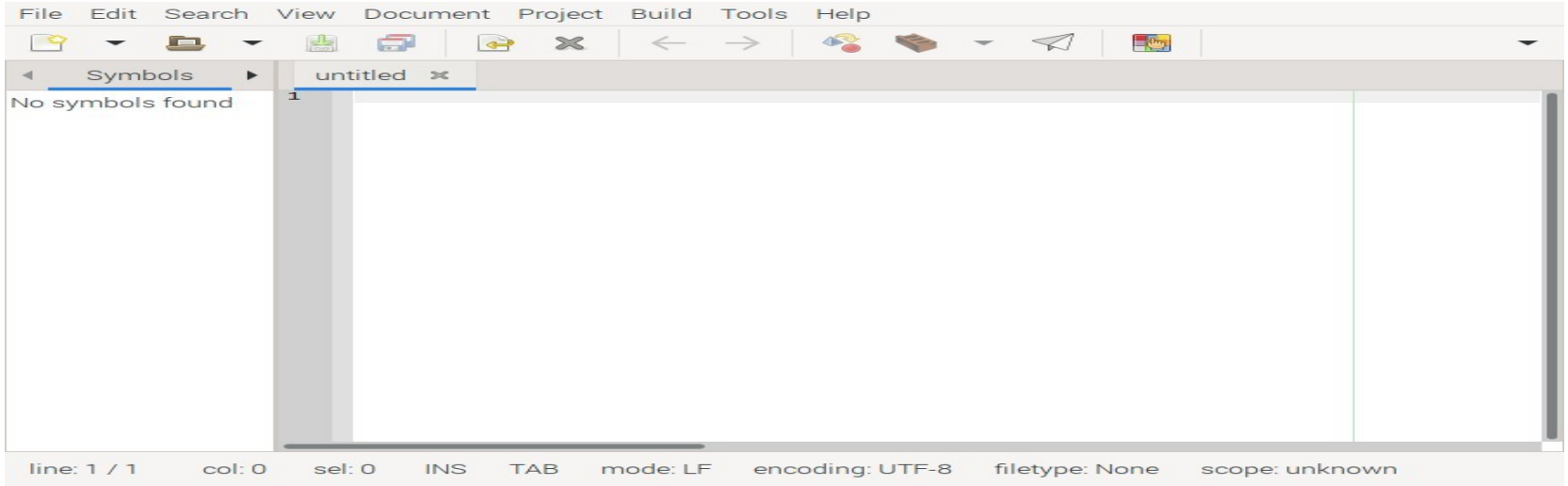
RPi CM4

RPi3B+

RPi4B

RPi3A


File Edit Tabs Help
devel@pi4-20:~/svd_rgb/src $ geany &
[1] 5418
devel@pi4-20:~/svd_rgb/src $


File Edit Search View Document Project Build Tools Help
Symbols
untitled
No symbols found
1
line: 1 / 1 col: 0 sel: 0 INS TAB mode: LF encoding: UTF-8 filetype: None scope: unknown

File Edit Search View Document Project Build Tools Help

Symbols

No symbols found

disp_S.m

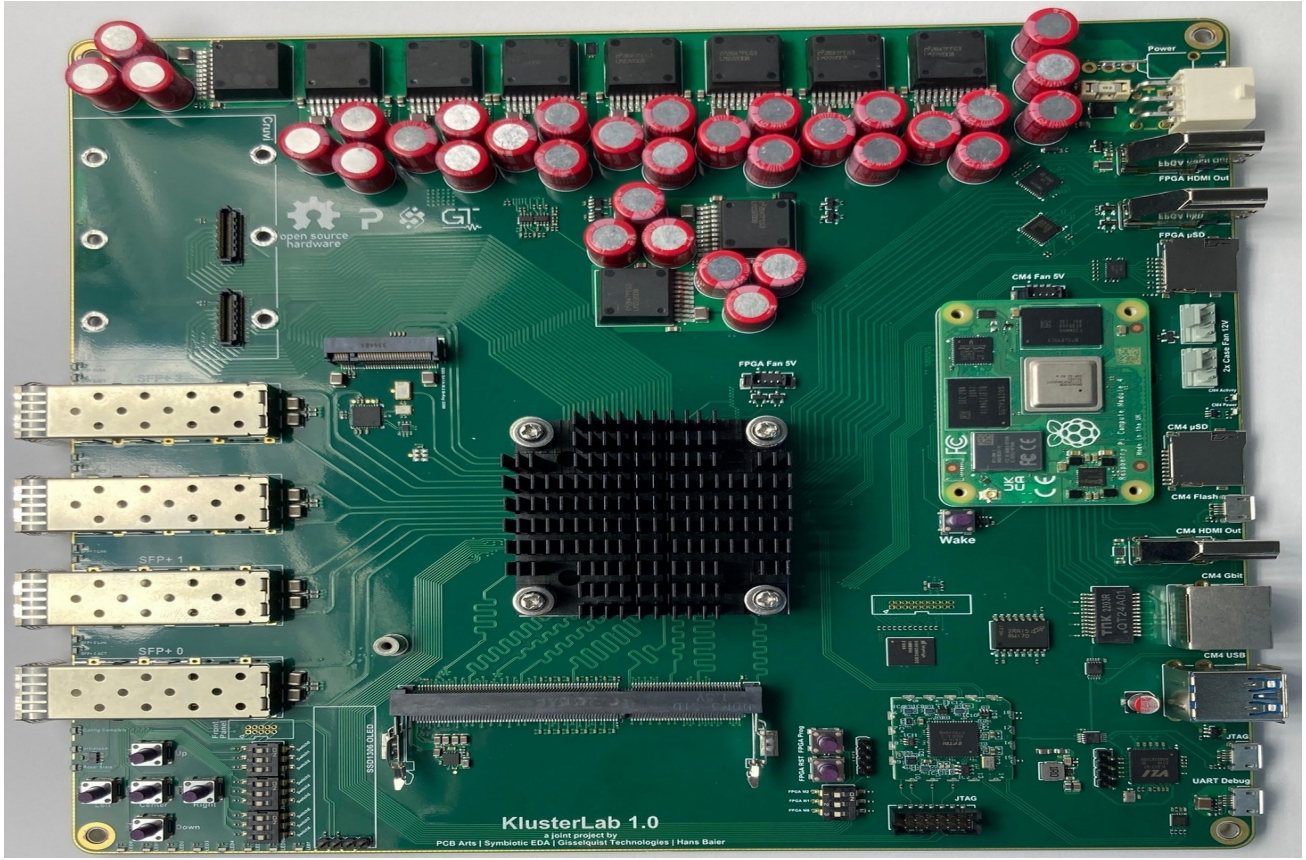
```
fid = fopen('Sblu.bin','r'); blu = fread(fid, 512, 'float');

fid = fopen('rcred.bin','r'); im1 = fread(fid, [512,inf], 'int32');
fid = fopen('rcgrn.bin','r'); im2 = fread(fid, [512,inf], 'int32');
fid = fopen('rcblu.bin','r'); im3 = fread(fid, [512,inf], 'int32');

aa(1:n)= red(1:n);
figure
stem(aa)
title("n S values red image from C img_svd")

figure;
imagesc(im1);
colorbar;
colormap 'gray';
title 'rcred svd from C'


bb(1:n)= grn(1:n);
figure
stem(bb)
title("n S value green image from C img_svd")

figure;
imagesc(im2);
colorbar;
colormap 'gray';
title 'rcgrn svd from C'

```

line: 67 / 101 col: 42 sel: 35 INS TAB mode: LF encoding: UTF-8 filetype: Matlab/Octave scope: unknown

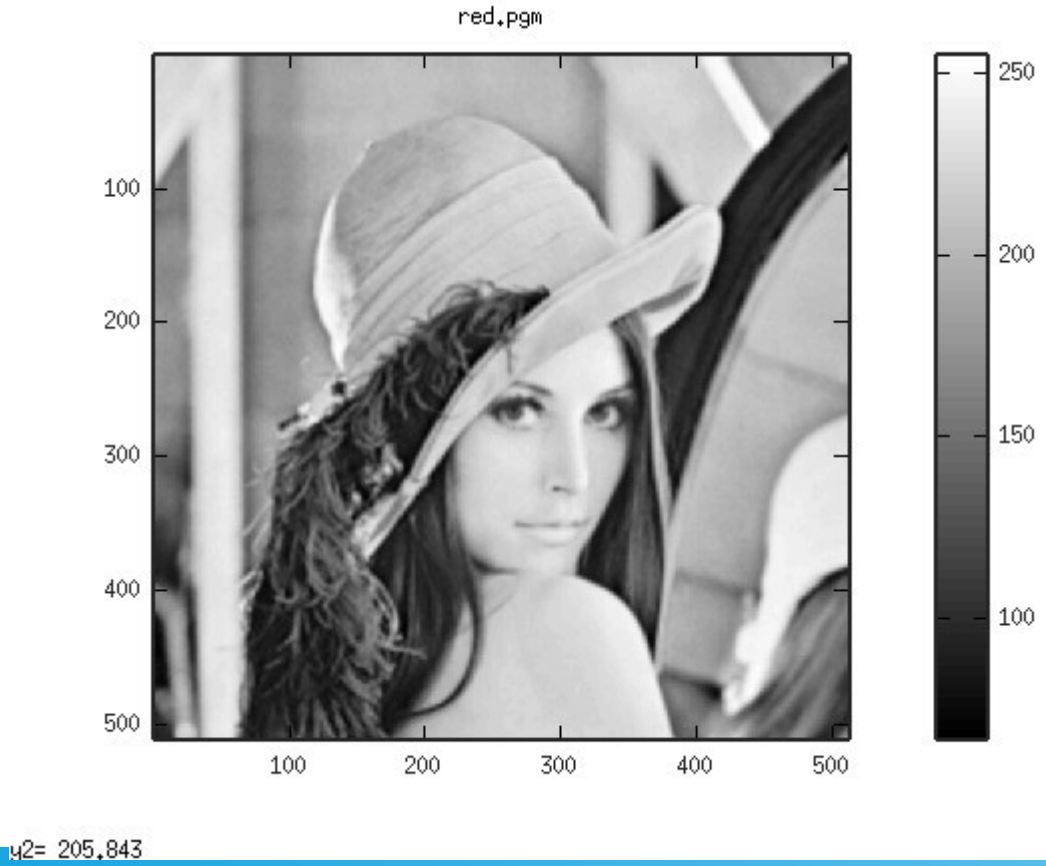
File Edit Tabs Help

```
devel@pi4-20:~/svd_rgb/src $ octave
GNU Octave, version 6.2.0
Copyright (C) 2021 The Octave Project Developers.
This is free software; see the source code for copying conditions.
There is ABSOLUTELY NO WARRANTY; not even for MERCHANTABILITY or
FITNESS FOR A PARTICULAR PURPOSE.  For details, type 'warranty'.

Octave was configured for "aarch64-unknown-linux-gnu".

Additional information about Octave is available at https://www.octave.org.

Please contribute if you find this software useful.
For more information, visit https://www.octave.org/get-involved.html

Read https://www.octave.org/bugs.html to learn how to submit bug reports.
For information about changes from previous versions, type 'news'.

octave:1> disp_S
octave:2>
```

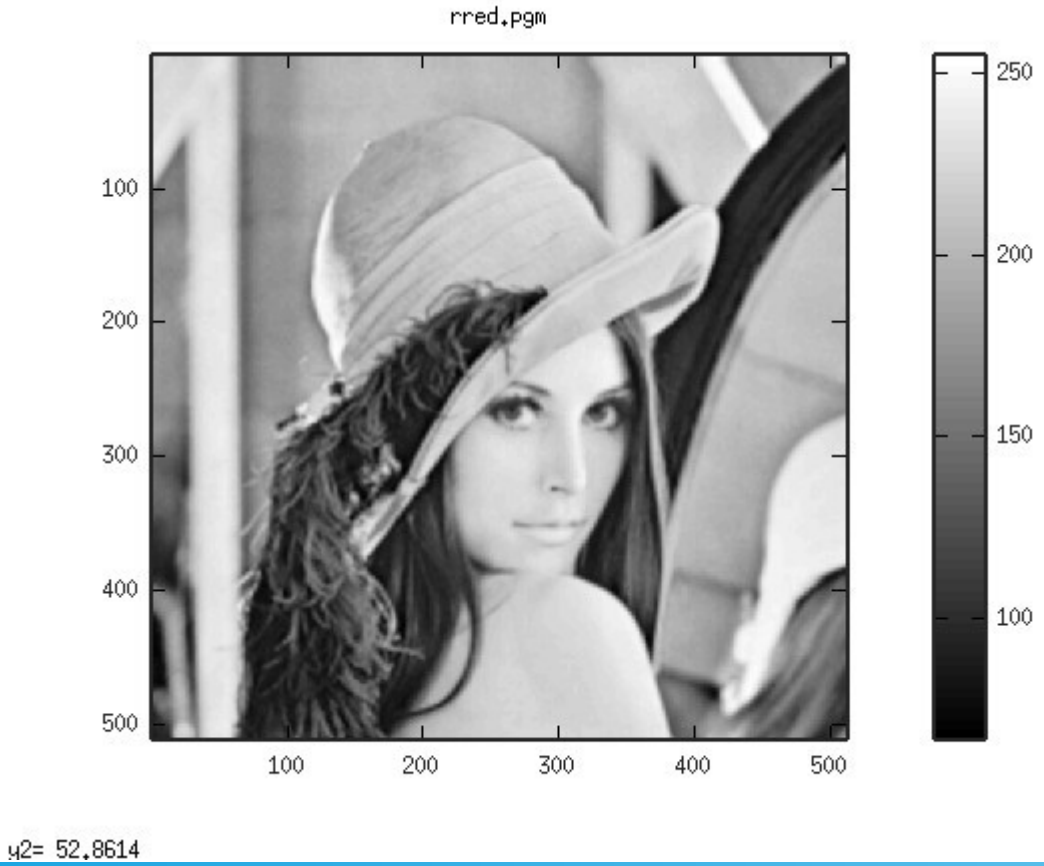
Rpi CM4 with FPGA Kintex-7 160T with 4 Ethernet 10Gb



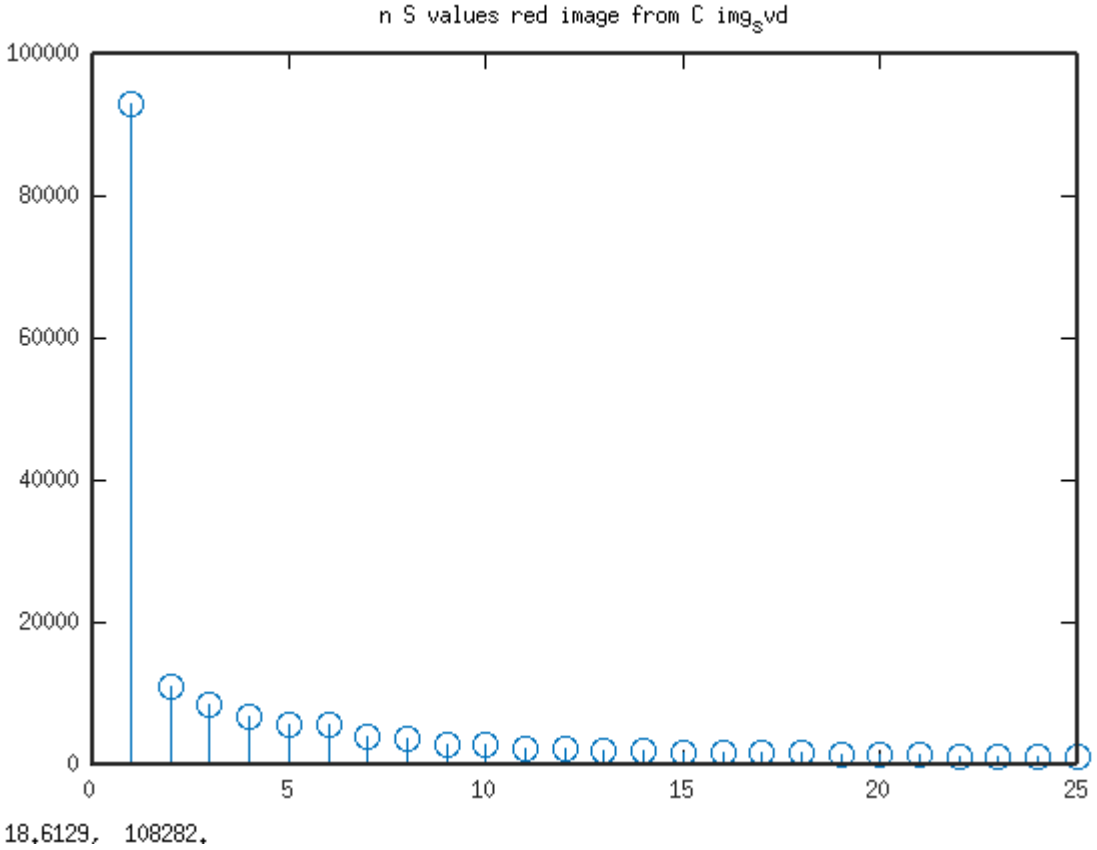
500 x 500 red subband lena



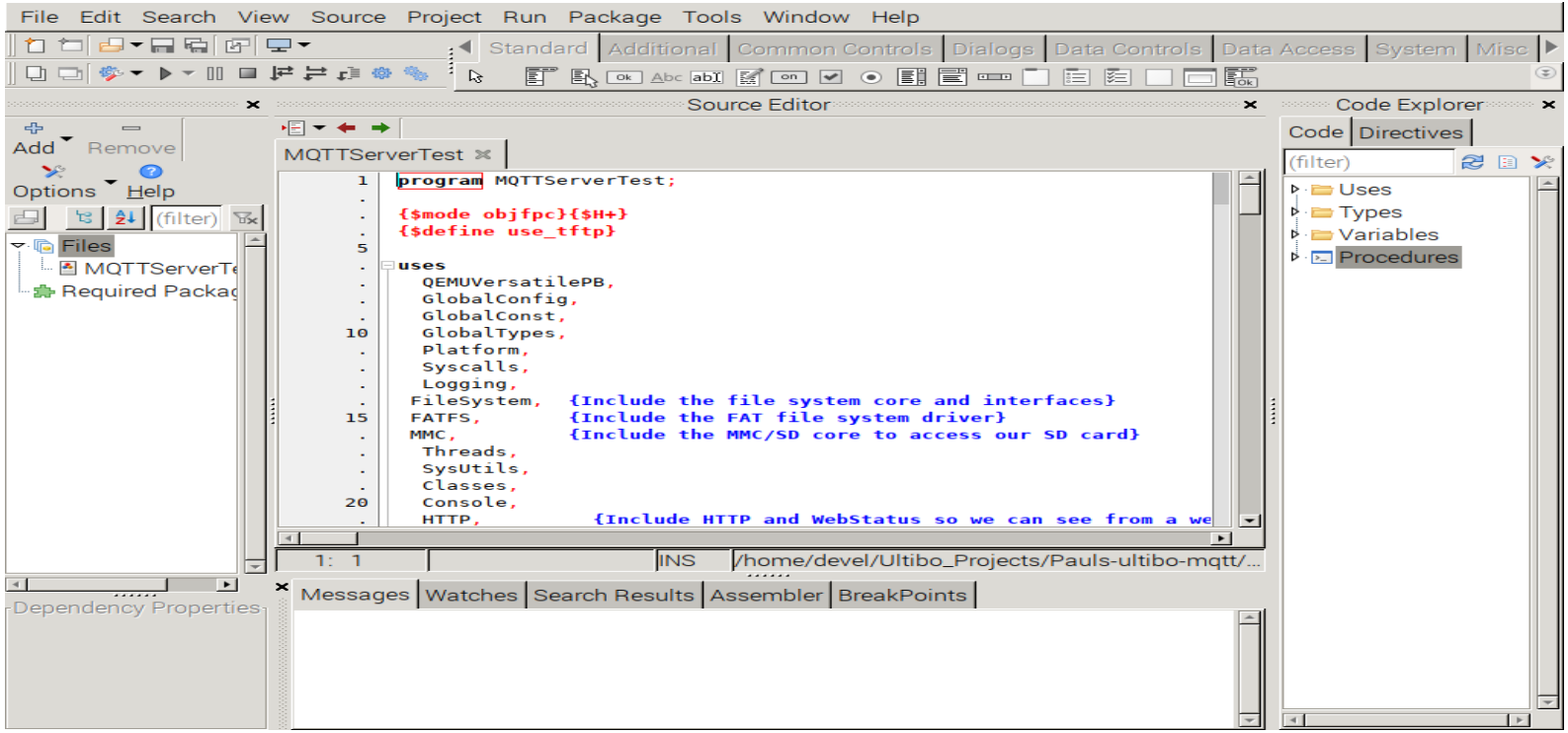
500 x 500 red subband lena reconstructed after perfoming KLT



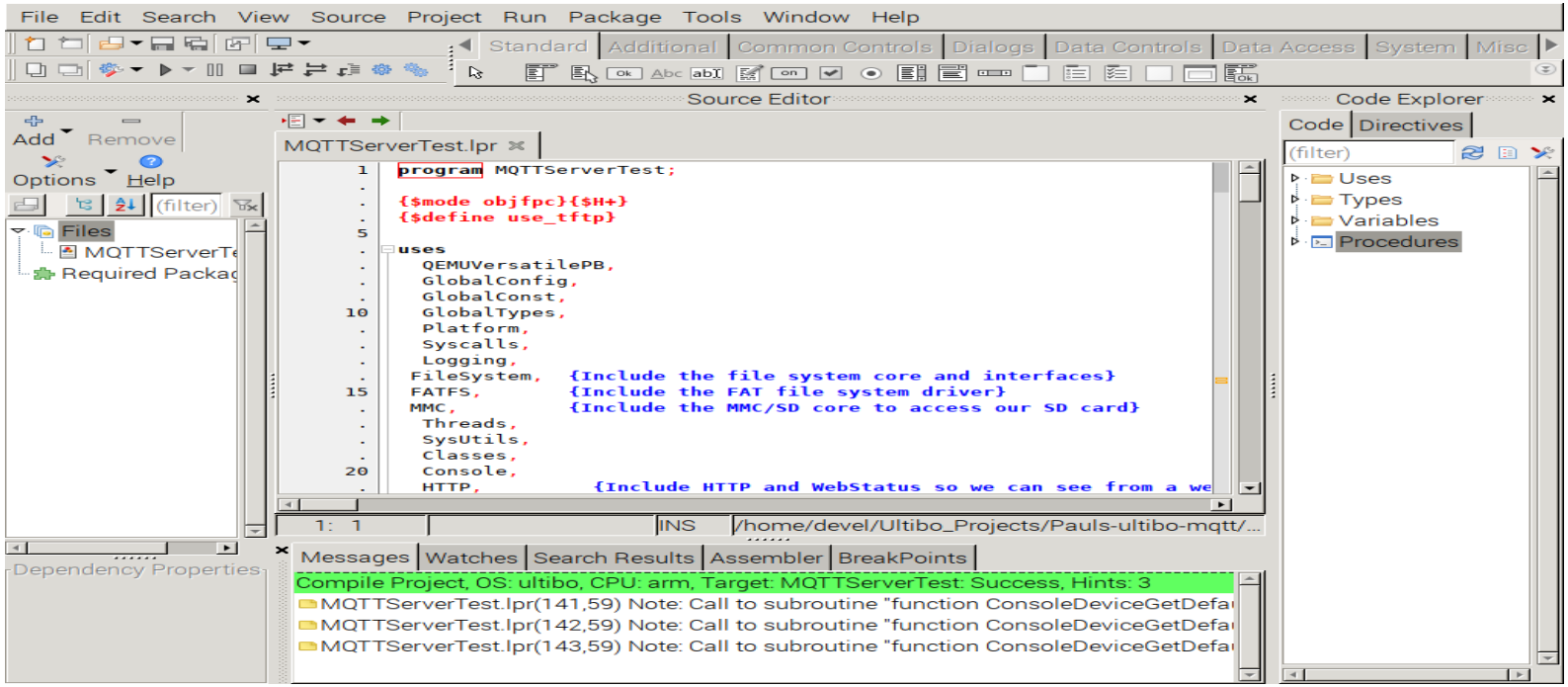
n S values red image from C img_svd



Goal: Accessing 1of 5 RPi4B from the Over the Internet with graphical displays

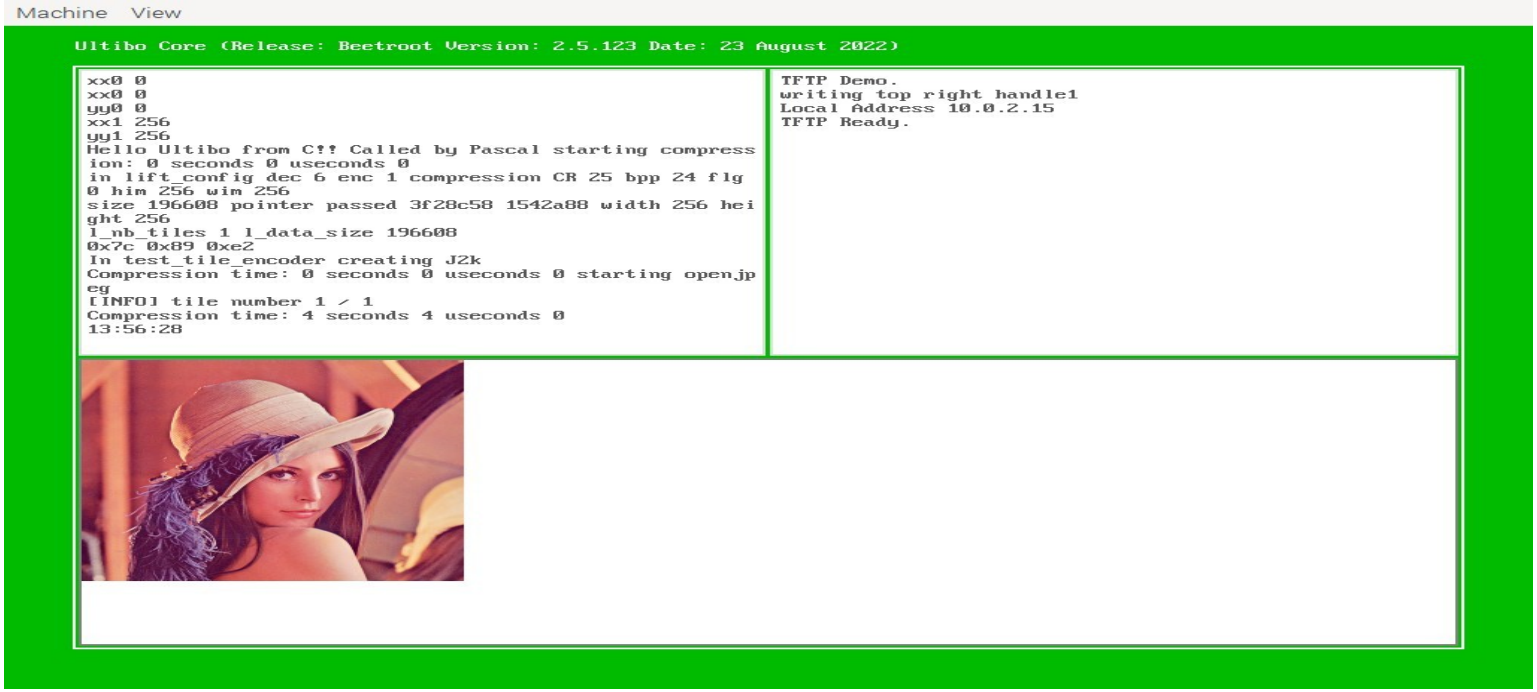


Goal: Accessing 1of 5 RPi4B from the Over the Internet with graphical displays



Goal: Accessing 1of 5 RPi4B from the Over the Internet with graphical displays

devel@pi4-20:~/Ultibo_Projects/jpeg2000/QEMU $ ./startqemu.s



ATT Router